

KALP SHAH

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Education

Indian Institute of Technology Gandhinagar

Jul. 2024 – Apr. 2028

B.Tech in Electrical Engineering with Minor in Computer Science and Engineering

GPA: 8.64/10 | Minor GPA: 10/10

Experience

AIResQ ClimSols

May 2026 – July 2026

Machine Learning Intern

- Engineered a **Computer Vision** pipeline for estimating depth of floods using CCTV Camera Networks
- Developed a **custom Camera and Radar device** for generation and validation of data in flood areas
- Implemented **single view metrology** algorithms for extracting features achieving **accuracy of $\pm 3\text{cm}$**

ABSix Robotics

May 2025 – July 2026

Software Development Intern

- Engineered a **cross-platform desktop application (Python, Qt)** with real-time robot-student communication via the **Dynamixel SDK** deployed across **15+ challenges** and actively used by **100+ students**
- Integrated a **RAG pipeline (LangChain + ChromaDB)** into the platform to query robotics concepts

Projects

Trade Matching Engine

Github

- Engineered a **C++ 26 limit-order matching engine** implementing price-time priority with FIFO Order execution
- Implemented efficient **order matching, cancellation and trade execution**, using stable **std::hive** iterators and direct order ID
- Benchmarked the matching engine across 200K+ orders using **hardware RDTSC timers**, achieving **12.5M+ orders/sec** throughput with **80 ns mean latency** (p50: 60 ns)

Belief Tracking for Countering LLM Negotiation Agents using Hidden Markov Models

Github

Supervisor: Prof. Manisha Padala

- Modeled LLM negotiation behavior** through a bargaining game simulation as a time-varying latent-state process using Hidden Markov Models
- Implemented HMM inference** using Forward-Backward, Viterbi, and Baum-Welch algorithms to track posterior beliefs and recover the opponent's hidden negotiation regime over time
- Developed belief-driven negotiation agents** and benchmarked them against no-belief, frequentist, and static-Bayesian baselines using negotiation payoff, regime-detection accuracy, and detection-lag metrics

Realtime Carbon Credit Market Analysis Platform

Github

Inter-IIT Technical Meet 14.0 – December 2025

- Architected a **full-stack real-time analytics platform** (Next.js, WebSocket, PostgreSQL) serving live carbon credit pricing, trend signals, and risk analysis to end users as part of an 8-person team
- Developed a **multi-agent RAG system** (LangChain, DeepSeek) generating context-aware recommendations
- Built a **data streaming pipeline** using **Kafka** and **Debezium** (CDC) to capture live **PostgreSQL** changes, cutting data freshness latency from **60 minutes to under 3 seconds**

Identifying Landing Locations on Mars using Classical Computer Vision

Research Poster

Supervisor: Prof. Shanmuganathan Raman

- Developed a CV pipeline to identify **safe landing zones on Mars** with **ISRO** under compute constraints
- Designed a modular system utilizing Sobel derivatives for **slope detection**, Fast Fourier Transforms for **texture analysis**, and the Maximal Rectangle Algorithm for **zone identification**
- Achieved a verifiable, low-power solution with **1.5 seconds runtime** on consumer-level hardware

Technical Skills

Languages: Python, C++, C, HTML/CSS, JavaScript, SQL

Frameworks: PyTorch, OpenCV, Langchain, scikit-learn, Selenium, Next.js

Cloud/DevOps: Azure, Linux, Docker, Git, PostgreSQL

Achievements

Codeforces Rating – Handle: *kalpshah1808* | **Expert:** 1746

Finalist at AMS Derive 2026 conducted by **Qube Research** and **Jane Street** at IIT Bombay

JEE Mains 2024 – All India Rank **3917** and JEE Advanced 2024 – All India Rank **6637**

Selected for **Barclays** BIIC Mentorship Program and ranked in top 5 for building FX Strategies